

Summarizing Posterior

UBCO MDS — DATA 582



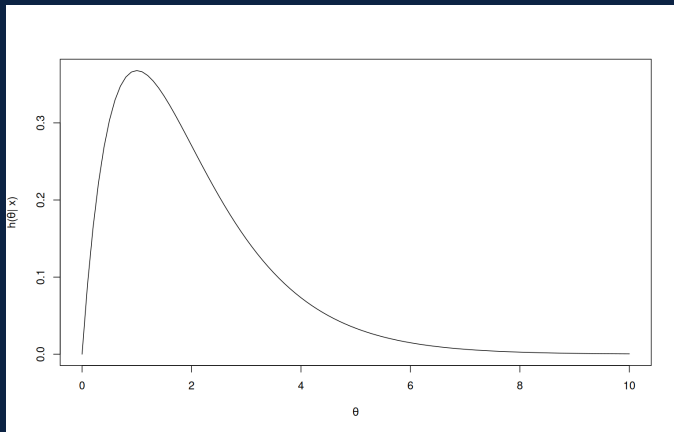


- ▶ **Bayesians** consider the path to inference by assuming variability in the **parameters** θ . Notably, they consider the observed data $\mathbf{x}$ FIXED.
- ▶ **Frequentists** consider the path to inference by assuming variability in the **observed data** $\mathbf{x}$. Notably, they consider the parameters θ FIXED.
- ▶ For Bayesian, we are seeking $h(\theta | \mathbf{x})$: the posterior distribution of θ .
- ▶ We've now got some experience finding $h(\theta | \mathbf{x})$, but what should we do with it?

Recall: Posterior Distribution



- Pretend we simply had $h(\theta | \mathbf{x})$ on hand and that θ is some super important parameter: maybe, the global temperature increase going forward from today that causes humanity's demise.



- What might you do with this info?



- ▶ A quick refresher on frequentist estimators. Let's assume $X \sim \text{Bernoulli}(p)$. Now, we know a natural estimator for p would be $\hat{p} = \frac{\sum x_i}{n}$.
- ▶ For some fixed p , if we consider repeated samples of size n for X , we find that
 - ▶ $E[\hat{p}] = p$
 - ▶ $\text{Var}[\hat{p}] = \frac{p(1-p)}{n}$
- ▶ Frequentist inference suggests that our 'best guess' for p is $\hat{p}$.
- ▶ Our 'best guess' at the variance of that estimator is the plug-in $\frac{\hat{p}(1-\hat{p})}{n}$...which we could then use for CIs.



- ▶ A point estimate in the Bayesian paradigm is arguably just as useful as a point estimate in the frequentist paradigm.
- ▶ Even if I assume θ is a random variable, I might still want to provide a 'best guess' at what its current (unobservable) state is.
- ▶ I see this as basically equivalent to the frequentist 'best guess' at the true, fixed (but still unobservable) state of θ .
- ▶ In frequentist inference, we define several objectives that we might wish a 'good' point estimate to have for the parameter θ
 - ▶ unbiased (or, at least, low bias)
 - ▶ low variance



- ▶ Since we have a generative posterior distribution for θ in the Bayesian paradigm, we simply have to decide what a good point estimate to choose would be.
- ▶ The most common choice is the **posterior mean**, which is simply the expected value of the posterior distribution.
- ▶ Mathematically, for a continuous posterior we would have

$$E[\theta] = \int \theta h(\theta | x) d\theta$$

- ▶ If our posterior is a recognizable distribution, this will often be extremely easy to find.



- ▶ Another common choice is the **posterior median**, which is the point where 50% of the distribution falls both above and below.
- ▶ As with measures of central tendency, this is sometimes preferred when the distribution has a lot of skewness.
- ▶ The **posterior mode** is sometimes used, which is the point at which the posterior is maximized.
- ▶ These are not always available in simple closed form, but generally simple to find through software.

Example: Bayes-ball



- ▶ Let $X_1, \dots, X_n$ represent a hit (1) or out (0) during 'at bats' for an MLB rookie we'll call Jacob.
- ▶ We'll assume each X_i is an independent Bernoulli trial with probability of success p .
- ▶ p could therefore be interpreted as Jacob's 'true' batting average
- ▶ Of course, $0 \leq p \leq 1$, where 0 would guarantee Jacob doesn't get a hit and 1 conversely ensures he gets a hit.

Example: Bayes-ball

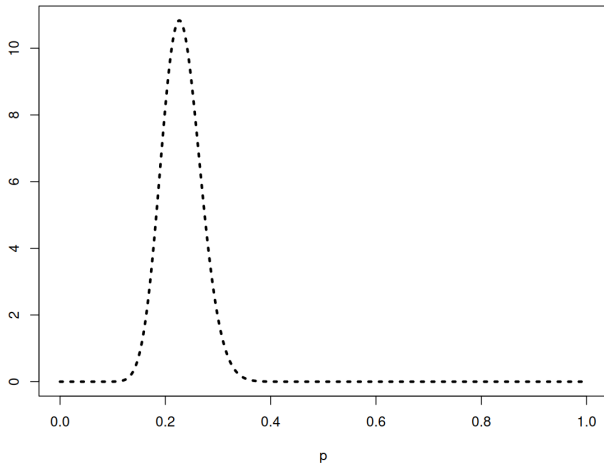


- ▶ An 'average' hitter tends to have a batting average around 0.25 for their career (1 in 4 at bats result in a hit), and a top tier hitter can end up around 0.33 (1 in 3 at bats).
- ▶ This suggests we have good information about what a p could/should look like for an MLB player before we ever see them in a game! We know
 - ▶ p is bounded $[0,1]$
 - ▶ p is unlikely to exceed 0.4 (nobody has achieved a career average this high)
 - ▶ p is unlikely to fall below 0.1 (they are in MLB, after all)

Example: Bayes-ball



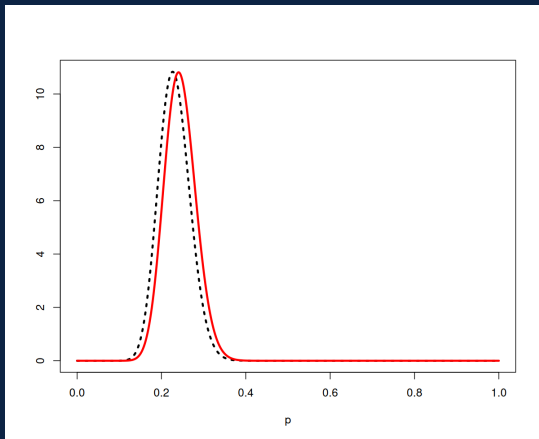
- So, perhaps a prior of $\text{Beta}(30, 100)$?



Example: Bayes-ball, First Game

- ▶ Now, suppose in Jacob's first game he gets 3 hits in 5 at bats.
- ▶ A Bayesian first computes the posterior to be

$$\text{Beta}(30 + \sum x_i, \quad 100 + n - \sum x_i) = \text{Beta}(33, 102)$$



Example: Bayes-ball, First Game



- ▶ The expected value for a $\text{Beta}(\alpha, \beta)$ is $\frac{\alpha}{\alpha + \beta}$, thus a (posterior mean) Bayesian point estimate for Jacob's batting average is

$$\hat{p}_b = \frac{33}{33 + 102} = 0.244$$

- ▶ A frequentist's best guess for Jacob's true batting average is

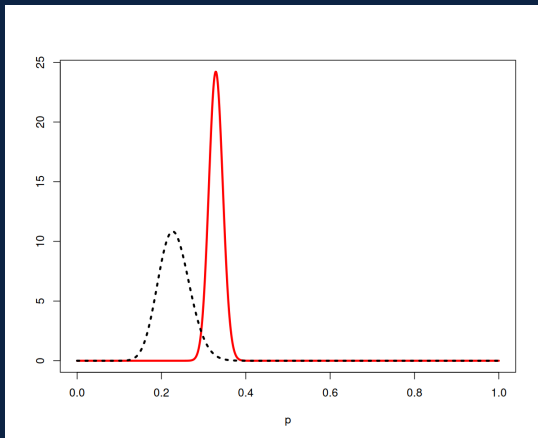
$$\hat{p}_f = \frac{\sum x_i}{n} = \frac{3}{5} = 0.6$$

- ▶ It is easy to argue that the Bayesian approach is more believable here (no player in history has come even close to batting at 0.6 for their career).
- ▶ The Bayesian estimator is therefore incorporating our prior belief that Jacob is extremely unlikely to be truly batting at that success rate.

Example: Bayes-ball, First Season

- ▶ Jacob's first season he gets 238 hits during 684 at bats.
- ▶ A Bayesian computes the posterior to be

$$\text{Beta}(30 + \sum x_i, 100 + n - \sum x_i) = \text{Beta}(268, 546)$$



Example: Bayes-ball, First Game



- ▶ The expected value for a $\text{Beta}(\alpha, \beta)$ is $\frac{\alpha}{\alpha + \beta}$, thus a (posterior mean) Bayesian point estimate for Jacob's batting average is

$$\hat{p}_b = \frac{268}{268 + 546} = 0.329$$

- ▶ A frequentist's best guess for Jacob's true batting average is

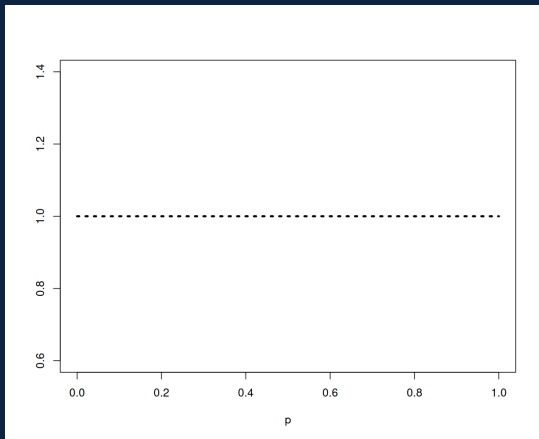
$$\hat{p}_f = \frac{\sum x_i}{n} = \frac{238}{684} = 0.348$$

- ▶ At this point, the high sample size is driving the Bayesian estimate enough to be approximately equivalent to the frequentist estimate.

Example: Bayes-ball, Uninformative Prior



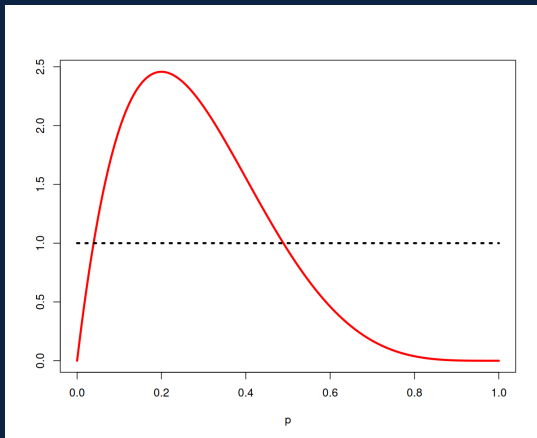
- What if we were completely unaware of what batting averages could/should look like in MLB, aside from knowing they are bounded $[0,1]$? Beta(1, 1)...



Example: Bayes-ball, First Game

- ▶ Now, suppose in Jacob's first game he gets 1 hit in 5 at bats.
- ▶ A Bayesian first computes the posterior to be

$$\text{Beta}(1 + \sum x_i, 1 + n - \sum x_i) = \text{Beta}(2, 5)$$



Example: Bayes-ball, First Game

- ▶ The expected value for a $\text{Beta}(\alpha, \beta)$ is $\frac{\alpha}{\alpha + \beta}$, thus a (posterior mean) Bayesian point estimate for Jacob's batting average is

$$\hat{p}_b = \frac{2}{2 + 5} = 0.286$$

- ▶ A frequentist's best guess for Jacob's true batting average is

$$\hat{p}_f = \frac{\sum x_i}{n} = \frac{1}{5} = 0.2$$

- ▶ Note that though these estimates are fairly close, the posterior mean estimate is pulled towards the expected value of the prior (0.5)
- ▶ It is therefore not the case that this 'uninformative' prior is completely allowing the data to speak for itself, at least when using the posterior mean to summarize...

Example: Bayes-ball, First Game



- ▶ If we switch to posterior mode though, note that the maximum point of a $\text{Beta}(\alpha, \beta)$ distribution can be computed via

$$\frac{\alpha - 1}{\alpha + \beta - 2}$$

- ▶ Thus, the posterior mode is

$$\hat{p}_m = \frac{2 - 1}{2 + 5 - 2} = \frac{1}{5} = 0.2$$

- ▶ Which is exactly equivalent to the frequentist point estimate!



- ▶ Of course, if point estimates are an interesting summary of a posterior distribution, so too would be variability measures
- ▶ We could consider the variance/standard deviation of the posterior distribution
- ▶ Even interquartile range, or other percentiles of the distribution, could help give us an understanding of what is happening with θ
- ▶ But, just like in the frequentist case, it may be more natural to incorporate both point estimates and variability into a range of values to guess for θ .



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